

Appendix: From the Sum-Zero of Fluctuations to Uncertainty

— The Uncertainty Relation as the Allocation of a Half-Bit

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Purpose

This appendix develops independently a single point compressed into a few lines in §6.3 (“the conservation law of the fluctuation”) of the related paper — namely that **the uncertainty relation in the observable space (constant product) is derived from the sum-zero of fluctuations in the fundamental-quantity N-space**. It adds no new claim to the parent paper; its aim is to make explicit a derivation whose direction is easily misread.

What is emphasized in particular is the **direction** of the derivation. Ordinarily the uncertainty relation $\delta_a \cdot \delta_b = \text{constant}$ (constant product) is given as a fundamental law. In this system it is a derived quantity, and what is placed as a definition is rather the **sum-zero** $\Delta_a + \Delta_b = 0$ in N-space. Sum-zero is the definition, constant product is the consequence — this direction is the heart of the matter.

This appendix is not a rigorous proof but an observation that makes explicit the structure of §6.3 of the parent paper. The magnitude of the fluctuation $\pm 1/2$ is treated as a posit, not a derivation (parent paper §6.2). No identification with physical quantities (position, momentum, etc.) is made. “Uncertainty” here is not in the sense of Bell’s theorem; it is limited to the formal structure of the conservation of a quantity allocated to a conjugate pair.

1. Setup: the fundamental quantity and the fluctuation

Following the related paper, the fundamental quantity is set as the logarithm of a squared quantity,

$$N \equiv \frac{1}{2} \log_2(\nu^2) = \log_2 \nu \quad (N \in \mathbb{Z}),$$

and the observable v is restored by the inverse map $v = 2^{\wedge\{N\}}$.

The conjugate pair a, b (corresponding to observables v, λ) has its product conserved:

$$ab = k \xrightarrow{\log_2} N_a + N_b = K \quad (K \equiv \log_2 k, \text{ constant}).$$

That is, in N -space the conjugate pair satisfies the **conservation of the sum** $N_a + N_b = K$. This is an additive identity and by itself generates no structure (note in §5 of the parent paper). Structure arises from the fluctuation riding on top of this sum.

2. The posit of the fluctuation: the half-bit $\pm 1/2$

Each fundamental quantity N has a fluctuation Δ . In this system we set it as

$$\Delta = \pm \frac{1}{2} \quad (\text{a half-bit; a posit; scale-invariant}).$$

This $\pm 1/2$ is a posit, not a derivation (parent paper §6.2: this system has no basis for constructing a standard deviation — individuation of observations, temporal order, persistence of value — and any computation smuggles in a hidden premise). Placing the fluctuation on the side of the fundamental quantity N rather than the observable v is because, when the fluctuation is held constant in N -space, it becomes scale-dependent in the observable space, which is natural as a fluctuation of integer counting.

It is called a “half-bit” because $\pm 1/2$ in the base-2 logarithmic space is half of one bit (the integer step of N). The base being 2 is by §3 and §4 of the parent paper (the binarity of the system).

3. Definition: the sum-zero of fluctuations

The sum of the conjugate pair is conserved ($N_a + N_b = K$, §1). Since K is a constant, the fluctuations Δ_a, Δ_b of the two fundamental quantities are constrained to preserve the sum. We place this as the **definition** of this system:

$$\boxed{\Delta_a + \Delta_b = 0}$$

That is, if one fluctuates by $+1/2$, the other necessarily fluctuates by $-1/2$. Which member of the conjugate pair the half-bit is allocated to is the degree of freedom of the fluctuation, and its total is always held at zero.

Figure 1: Sum-zero in N-space. The conjugate pair (N_a, N_b) lies on the line $N_a + N_b = K$; the half-bit fluctuation shifts one by $-1/2$ and the other by $+1/2$. The point slides along the line and the sum is held at K .

This sum-zero is a rephrasing of the requirement that the sum of the conjugate pair $N_a + N_b = K$ be conserved even when fluctuations are included. Writing the fundamental quantities with fluctuations added as

$$\tilde{N}_a = N_a + \Delta_a, \quad \tilde{N}_b = N_b + \Delta_b,$$

we have

$$\tilde{N}_a + \tilde{N}_b = (N_a + N_b) + (\Delta_a + \Delta_b) = K + 0 = K,$$

so the sum is held exactly at K even under fluctuations.

Geometrically, the conjugate pair (N_a, N_b) is a point on the line $N_a + N_b = K$, and the fluctuation $\Delta_a = -1/2$, $\Delta_b = +1/2$ (or vice versa) is a slide along this line. The point stays on the line and the sum is conserved (Figure 1).

4. Derivation: constant product (the uncertainty relation)

We map the sum-zero (§3) back to the observable space. The observables are restored by $a = 2^{N_a}$, $b = 2^{N_b}$:

$$ab = 2^{\tilde{N}_a + \tilde{N}_b} = 2^{K+0} = 2^K \quad (\text{constant}).$$

That is, under the sum-zero constraint, the product ab of the observables is held constant at 2^K regardless of the fluctuation.

Consider the fluctuation of each observable. Around the central value $a_0 = 2^{N_a}$,

$$a = 2^{N_a + \Delta_a} = a_0 \cdot 2^{\Delta_a}, \quad b = b_0 \cdot 2^{\Delta_b}.$$

Their product is

$$ab = a_0 b_0 \cdot 2^{\Delta_a + \Delta_b} = a_0 b_0 \cdot 2^0 = a_0 b_0,$$

so however the fluctuations Δ_a , Δ_b are allocated, the product is fixed at the product of the central values $a_0 b_0$. Under sum-zero, the half-bit remains irreducibly as $\pm 1/2$ on each fundamental

Figure 2: Constant product in the observable space. The conservation relation is the hyperbola $a \cdot b = 2^K$; tighten one and the other widens. This is the image, under the inverse logarithmic map, of the sum-zero in N-space (Figure 1).

quantity, and its sign is anti-correlated (if one is $+1/2$ the other is $-1/2$). In the observables, if one is $\sqrt{2}$ times, the other is $1/\sqrt{2}$ times, and the product is fixed at $a_0 b_0$.

This is the form of the **uncertainty relation** in the observable space: the half-bit remains on each observable as a floor that cannot be removed, and the product of the conjugate pair $a \cdot b$ is fixed at $a_0 b_0 (=2^K)$. Note that what is preserved is the **product of the observables** $a \cdot b$, not the **product of the fluctuations** $\delta_a \cdot \delta_b$ (the latter depends on the allocation Δ and is not constant). In the squeeze where Δ is taken continuous (next, Figure 2), this appears as the trade-off “tighten one and the other widens,” and the fixed half-bit $\pm 1/2$ is its minimal unit. This constant product is a consequence derived from the sum-zero in N-space (§3, the definition), not a law posited independently.

The minimal unit of uncertainty corresponds to the posited half-bit $\pm 1/2$. Note that a single fluctuation $\pm 1/2$ and the conserved quantity appearing as the product of two fluctuations ($\delta_a \delta_b = 1/2$, Axiom 4 of the parent paper) are not the same quantity, but they are placed in correspondence as quantities belonging to the same side — area / the symplectic form ω (the imaginary part of the complex inner product).

In the observable space, this conservation relation appears as the hyperbola $a \cdot b = 2^K$. Tightening one (making it small) widens the other, and the two move along the hyperbola while the product is held constant (Figure 2). The line in N-space (sum-zero) corresponds, via the inverse logarithmic map, to the hyperbola in the observable space (constant product).

5. Summary of the structure

Space	Quantity	Fluctuation	Constraint
Fundamental N (definition side)	$N = \log_2 v$ (integer)	$\Delta = \pm 1/2$ (half-bit, posit)	$\Delta_a + \Delta_b = 0$ (sum-zero, definition)
Observable v (derived side)	$v = 2^N$	$\delta = v_0(2^\Delta - 1)$ (scale-dependent)	$a \cdot b = 2^K$ (the product of observables is constant, derived)

The definition is the **sum-zero** in N-space; the derivation is the **constant product** in the observable

space. The body of “uncertainty” is the allocation structure “to which member of the conjugate pair the half-bit $\pm 1/2$ is allocated, with the total conserved,” and the constant product is its shadow in the observable space.

Multiplication (constant product) and addition (sum-zero) are connected by the logarithmic map $N = \log_2 v$. Uncertainty, which appears as a product in the observable space, becomes the simple conservation law of sum-zero in the fundamental-quantity space — this is one consequence of taking the logarithm as the fundamental quantity.

6. Limitations (reading notes)

This appendix makes explicit the structure of §6.3 of the parent paper and adds no new claim. The following are stated.

1. **$\pm 1/2$ is a posit, not a derivation.** As in §6.2 of the parent paper, this system has no basis for constructing a standard deviation. The derivation of this appendix (sum-zero $\rightarrow$ constant product) presupposes this posit. If the posit fails, the consequence changes.
2. **Sum-zero $\Delta_a + \Delta_b = 0$ is placed as a definition.** It is a rephrasing of the requirement that the conservation of the sum of the conjugate pair ($N_a + N_b = K$) hold even under fluctuations, and is not derived from a more fundamental principle.
3. **Uncertainty here is not in the sense of Bell’s theorem.** It is limited to the formal structure of the conservation of a quantity allocated to a conjugate pair, and is distinct from the EPR/Bell no-hidden-variables result.
4. **No physical identification is made.** The physical names of the conjugate pair a, b (observables v, λ) — position, momentum, time, energy, etc. — are anonymous labels, and this appendix makes no identification with a particular physical quantity.
5. **It is distinct from the integer-step fluctuation of §6.1.** The “the interior integer quantity fluctuates in integer steps (± 1)” of §6.1 of the parent paper and the half-bit fluctuation $\Delta = \pm 1/2$ placed in N -space here are separate posits, and no relation (identity, derivation, approximation) is claimed between them.
6. **What is conserved is the product of the observables, not the product of the fluctuations.** What sum-zero guarantees is the product of the observables $a \cdot b = 2^K$, not the product of the fluctuations $\delta_a \cdot \delta_b$ (the latter depends on the allocation and is not constant). Also, what the sum-zero of the fixed half-bit gives is “an irreducible $\pm 1/2$ floor remaining on each observable + anti-correlation of signs”; this is a different aspect from the squeeze where Δ is taken continuous (the trade-off on the hyperbola, “tighten one and the other widens”). The fixed half-bit $\pm 1/2$ is the minimal unit of that squeeze. The two are not identified.

Reference

- Kihara, N. “On the Logarithmic Map of Scale and the Separation of Scale and Quantization in a High-Symmetry Limit — A Limited Verification Report from a Minimal Axiom System,” §6.2, §6.3. Concept DOI: [10.5281/zenodo.20740841](https://doi.org/10.5281/zenodo.20740841).